Zhengyu Hu, Ph.D.

Civil & Environmental Engineering, National University of Singapore

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Research Interests

Wave-structure-seabed interactions;
Breaking wave impact;
Sediment transport and scour;
Nature-based coastal protection.

Professional Experience

Oct 2024 – Present

Research Fellow, **Project Manager**, National University of Singapore.
– Manage SGD 2.5 million National Research Foundation project on climate-driven coastal hazards and protection.
– Conduct large-scale CFD simulations to support regional coastal hazard analysis under extreme forcing.

Education

Aug 2021 – Feb 2025

Ph.D., National University of Singapore. Advisor: Asst. Prof. Yuzhu Pearl Li.
Thesis title: *Fully Coupled Model for Wave–structure–seabed Interaction in Coastal Engineering.*

Sep 2018 – Jun 2021

M.Eng, Zhejiang University. Advisor: Prof. Fang He.
Thesis title: *Laboratory Study on Characteristics of Wave Attenuation by Mangroves and Phragmites Australis*.

Publications

Journal Articles

1. **Hu**, **Z.**, Chen, W., and Li, Y. P., "Sediment rheology governs wave damping over muddy shelves," *Communications Earth & Environment*, accepted, 2026.
2. **Hu**, **Z.**, Yue, L., Fuhrman, D., and Li, Y. P., "The role of dynamic seepage in surf and swash zone sediment transport and morphodynamics," *Journal of Fluid Mechanics*, under review, 2026.
3. Pan, S., Zhang, L., Zhao, X., and **Hu**, **Z.**, "Fully coupled two-dimensional numerical simulation of wave interaction with an elastic wall in a narrow gap," *Physics of Fluids*, under review, 2026.
4. Zhou, Y., **Hu**, **Z.**, Wang, P. C., and Li, Y. P., "The efficacy of green and gray coastal structures in storm surge mitigation," *Coastal Engineering*, vol. 207, p. 104 972, 2026. DOI: 10.1016/j.coastaleng.2026.104972.
5. **Hu**, **Z.** and Li, Y. P., "Breaking wave impacts on an elastic plate," *Journal of Fluid Mechanics*, vol. 1015, A54, 2025. DOI: 10.1017/jfm.2025.10397.
6. **Hu**, **Z.** and Li, Y. P., "Full-scale tsunami-induced scour around a circular pile with three-dimensional seepage," *Coastal Engineering*, vol. 197, p. 104 676, 2025. DOI: 10.1016/j.coastaleng.2024.104676.
7. **Hu**, **Z.**, Qi, W.-G., and Li, Y. P., "The role of dynamic seepage response in sediment transport and tsunami-induced scour," *Journal of Geophysical Research: Oceans*, vol. 130, no. 3, e2024JC021084, 2025. DOI: 10.1029/2024JC021084.

8. Tavakoli, S., Singh, M., Hosseinzadeh, S., **Hu**, **Z.**, and et al., "A review of flexible fluid-structure interactions in the ocean: Progress, challenges, and future directions," *Ocean Engineering*, vol. 342, p. 122 545, 2025. DOI: 10.1016/j.oceaneng.2025.122545.

9. **Hu**, **Z.**, Huang, L., and Li, Y., "Fully-coupled hydroelastic modeling of a deformable wall in waves," *Coastal Engineering*, vol. 179, p. 104 245, 2023. DOI: 10.1016/j.coastaleng.2022.104245.

10. **Hu**, **Z.** and Li, Y., "Two-dimensional simulations of large-scale violent breaking wave impacts on a flexible wall," *Coastal Engineering*, vol. 185, p. 104 370, 2023. DOI: 10.1016/j.coastaleng.2023.104370.

Conference Proceedings (selected)

1. **Hu**, **Z.** and Li, Y., "Experimental study of the breaking wave impact on rigid and elastic plates," in *Proceedings of the 38th International Conference on Coastal Engineering*, Rome, Italy, 2025, structures.11. DOI: 10.9753/icce.v38.structures.11.

2. **Hu**, **Z.** and Li, Y., "Numerical study on the interaction between periodic waves and a flexible wall," in *Proceedings of the 37th International Conference on Coastal Engineering*, Sydney, Australia, 2023, structures.3. DOI: 10.9753/icce.v37.structures.3.

Patents (selected)

1. He, F., **Hu**, **Z.**, et al., "A coral seabed cultivation device and method suitable for harsh marine conditions," Patent CN110269005A, Granted, 2022.

2. He, F., **Hu**, **Z.**, et al., "A hollow cylindrical submarine coral cultivation device and method," Patent CN110269006A, Granted, 2022.

3. He, F., **Hu**, **Z.**, et al., "A submarine coral cultivation device that uses tidal energy to generate electricity," Patent CN111165393A, Granted, 2021.

4. He, F., **Hu**, **Z.**, et al., "A wave-powered floating box-type coral seabed cultivation device that facilitates coral growth," Patent CN110338115A, Granted, 2021.

5. He, F., **Hu**, **Z.**, et al., "A test device and method for simulating sediment scour and achieving automatic levelling based on electromagnetic induction," Patent CN109323847A, 2020.

Research Projects

Aug 2024 – Present
- **Project manager**, National Research Foundation, Singapore, SGD 2.5 million. Project: *Model development and quantifications of coupled near-shore processes impacted by climate change.*

Dec 2021 – Present
- **Researcher**, Ministry of Education, Singapore, SGD 0.4 million. Project: *On the mechanisms of wave-structure-soil interaction in coastal regions.*

Oct 2019 – Sep 2020
- **Principal investigator**, Zhejiang Provincial Department of Education, China. Project: *Research on the mechanism, function, and optimised application of coastal vegetation for ecological disaster reduction.*
- **Researcher**, Ministry of Natural Resources, China, CNY 0.6 million. Project: *Research and application of marine hazard reduction evaluation and establishment optimisation technology of typical coastal wetlands.*

Oct 2018 – Sep 2019
- **Researcher**, Ministry of Natural Resources, China, CNY 0.4 million. Project: *Study on the damage mechanism and the risk early warning method of marine dynamic hazard on coral reefs.*

Conference Presentations

4 – 7 Nov 2025 — **12th International Conference on Scour and Erosion (ICSE12)**, China.
Topic: *Full-scale tsunami scour with three-dimensional seepage flows*

22 – 27 Jun 2025 — **41st IAHR World Congress (IAHR2025)**, Singapore.
Topic: *Impact of seepage flows on tsunami-induced sediment transport and scour*

8 – 14 Sep 2024 — **38th International Conference on Coastal Engineering**, Rome, Italy.
Topic: *Experimental study of breaking wave impacts on rigid and elastic plates*

18 – 22 Jun 2024 — **Singapore International Water Week (SIWW 2024)**, Singapore
Topic: *An assessment of coastal protection effectiveness with soft structures*

9 – 14 Jun 2024 — **43rd International Conference on Ocean, Offshore and Arctic Engineering**, Singapore
Topic: *Scour assessment and prediction around jack-up footings in wave-current conditions*

1 – 4 Aug 2023 — **Asia Oceania Geosciences Society 2023**
Topic: *A coupled numerical model of tsunami-induced scour and seepage around a pipeline*

4 – 9 Dec 2022 — **37th International Conference on Coastal Engineering**, Sydney, Australia.
Topic: *Numerical study on the interaction between periodic waves and a flexible wall*

Teaching & Mentoring

Aug 2023 – Nov 2024 — **Teaching Assistant**, Sediment Transport & Coastal Protection, NUS
- Led lab tutorials to about 40 graduate students across 3 semesters.

Aug 2022 – Nov 2024 — **Teaching Assistant**, Fluid Mechanics, NUS
- Led lab and assignment tutorials to about 250 undergraduate students across 4 semesters.

Dec 2021 – Nov 2025 — **Co-supervisor**, Undergraduate/MSc Final Year Project, NUS
- Co-supervise the Final Year Project of 6 undergraduate and 4 MSc students.

Mar 2019 – Jun 2020 — **Teaching Assistant**, Port and Coastal Hydraulic Structures, ZJU
- Led lab tutorials to over 120 undergraduate students across 3 semesters.

Dec 2018 – Jun 2020 — **Co-supervisor**, Undergraduate Final Year Project, ZJU
- Co-supervise the Final Year Project of 3 undergraduate students.

Prizes and Awards

2025 — **Chinese Government Award for Outstanding Self-Financed Students Abroad**.
China Scholarship Council, USD 6,000.

Excellence in Reviewing.
Ocean Engineering, Elsevier.

2021 — **NUS Research Scholarship**.
National University of Singapore, SGD 80,000/year.

2018 — **Academic Scholarship**.
Zhejiang University, CNY 8,000/year.

Professional Service

Reviewer — *Coastal Engineering*, *Ocean Engineering*, *Physics of Fluids*, *Journal of Waterway, Port, Coastal, and Ocean Engineering*, *Fluids*, *Mathematics*, *Applied sciences*, etc. (85+ reviews).

Session Chair — 45th International Conference on Ocean, Offshore and Arctic Engineering (OMAE).